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Exploring Multi-Directional 3D Printing for Enclosures on Earth and Beyond: Sensitivity Analysis of Buildability

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ABSTRACT

Architecture in space demands innovative solutions to address the challenges of constructing robust, airtight habitats on extraterrestrial surfaces. Additive manufacturing (AM) has emerged as a promising technology for building habitats on celestial bodies, such as Mars, due to its adaptability and potential for automated construction. While 3D printing techniques on Earth have proven effective for building components, challenges persist in printing enclosures and the utilization of various mixed fabrication methods for roof construction. The study aims to lower the printable overhang angle, addressing restrictions posed by a 60-degree limit in horizontal or corbelling slicing techniques. Inspired by historical bricklaying, the research explores multi-directional printing with corbelling, radial, and inclined slicing techniques. The paper focuses on a sensitivity analysis study aimed at understanding the impact of geometric variables on the printability of barrel vaults using combined horizontal and inclined slicing techniques. A simulator of structural behavior during printing developed previously is used to predict the number of printed layers until collapse. Results suggest that it is possible to maintain the number of printable layers while lowering the overhang angle. While these results still need to be validated through printing experiments, they constitute a stepstone in the development of a design optimization tool to assist in the design of 3D printed enclosures.

INTRODUCTION

3D printing in space habitat construction is an innovative approach that addresses the challenges of building structures in harsh environments. The teams who participated in NASA's competition to design a 3D-printed habitat for Mars aimed to advance additive construction for sustainable housing on Earth, Moon, Mars, and beyond (Kowsky 2020). This innovative approach can be used to construct housing wherever affordable housing is needed and access to conventional building materials and skills are limited. Penn State secured second place in the competition with the enclosed habitat design illustrated in Figure 1 (Muthumanickam et al. 2020). While this habitat was successfully 3D printed without formwork, this achievement was contingent on the geometry having an overhang angle exceeding 60°, given the limitations of the corbeling techniques used, which involves printing with a vertical nozzle (Duarte et al. 2021). As a result, the enclosure shape needed to have a pointed arch to be printed without formwork (Figure 2). Due to this limitation, the use of 3DCP to print building components without formwork has been mainly restricted to walls. To enable lower structures and greater design flexibility, achieving a lower overhang angle is required. This study explores the hypothesis that by combining different slicing and multidirectional printing techniques, it is possible to lower the overhang angle.

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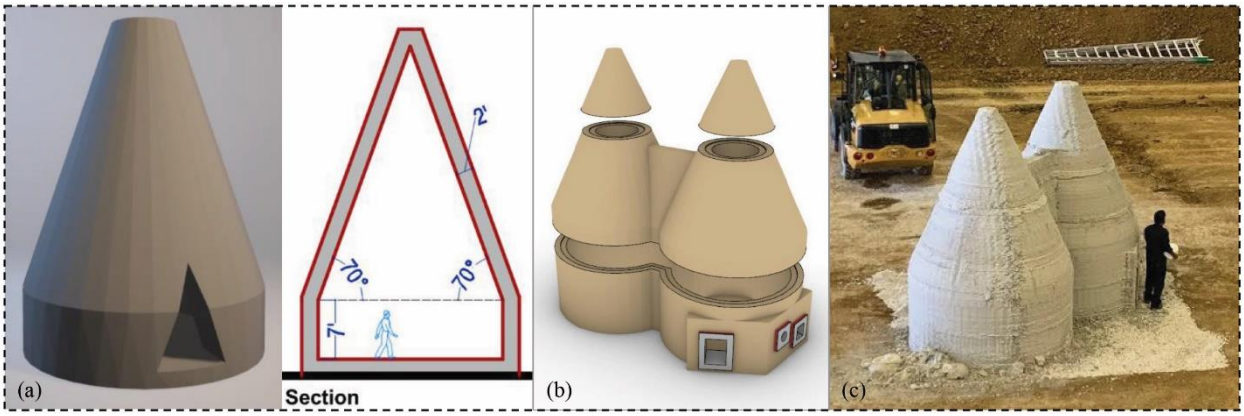

Figure 1. Design of NASA's 3D-Printed Habitat Challenge by Pennsylvania State University (Muthumanickam et al. 2020).

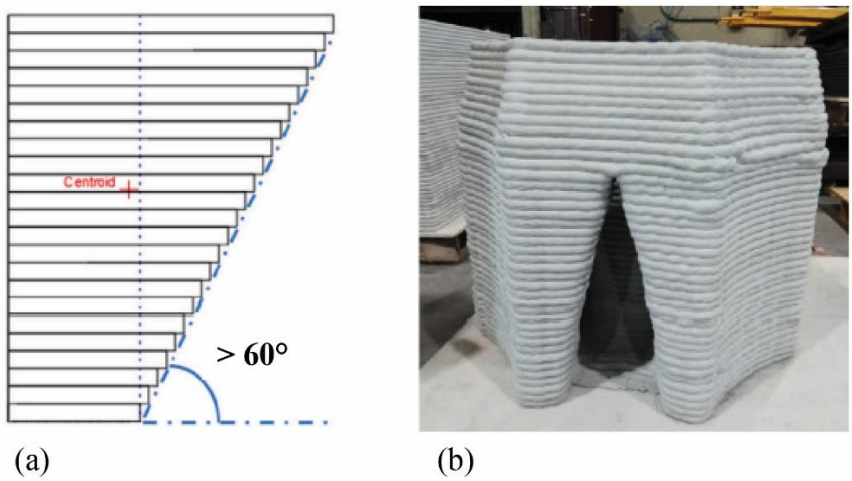

Figure 2. Overhang angle of 3DCP; (a) front view of layer deposition with a maximum overhang of 60°, and (b) self-supporting pointed arch (Duarte et al. 2021).

Formwork is a vital component of traditional concrete construction, influencing labor requirements, quality, timelines, and overall expenses, making it the most significant cost factor in such projects (Antony et al. 2014). Historical construction techniques for spanning structures offer various fabrication ideas without relying on formwork, particularly evident in stone, adobe, and masonry constructions. The utilization of diverse bricklaying patterns in self-standing spanning structures reflects the creativity and expertise of ancient architects and builders. While the prevalent slicing technique is corbelling, which involves horizontal slicing, other slicing methods include radial and inclined slicing (Figure 3) (Carneau et al. 2020). The implementation of these techniques through multi-directional printing can reduce the overhang angle without the need for formwork (Tabassum et al. 2023). To combine different slicing techniques in the same print, it is necessary to develop decomposition techniques to break down complex shapes into smaller, more manageable components for analysis and fabrication. Shape decomposition allows users to generate multiple designs while adapting to structural and fabrication constraints for 3D

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printing (Motamedi et al. 2022). Studies show that shape decomposition methods for multi-directional 3D printing can lead to effective additive manufacturing strategies and significantly reduce the need for support in certain geometries (Wu et al., 2019).

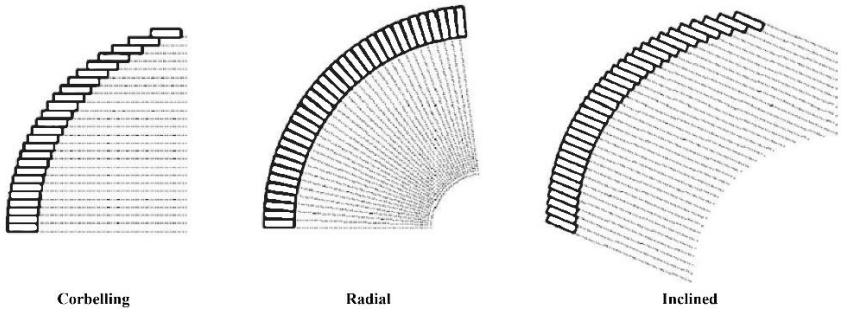

Figure 3. Different types of slicing techniques: horizontal or corbelling, radial, and inclined (Carneau et al. 2020).

In the realm of 3D printed concrete, the early-age properties of the material during printing determine the printability and shape of the structure. The buildability of freshly printed concrete is influenced by several factors, including rheology, stress-strain behavior, and time-dependent behavior. However, there is still limited research on the mechanical properties of 3D printed concrete. Understanding early-stage mechanical properties is crucial for designing and simulating the structural behavior of concrete during printing, especially in predicting failure modes such as plastic collapse, elastic buckling, and tensile failure. This paper uses the algorithm proposed by Duarte et al. 2023 to simulate structural behavior during printing to assess the feasibility of multi-directional printing strategies.

OVERVIEW OF THE COMPREHENSIVE RESEARCH

The primary research aim is to decrease the current overhang angle constraints in 3D concrete printing to facilitate the printing of enclosures without formwork using multi-directional printing. The goal is to develop a design methodology that guides designers through toolpath design, numerical modeling, and fabrication stages. The research involves a series of experiments to minimize the printing angle of a barrel vault using combined slicing techniques and multi-directional printing strategies. The comprehensive research of this study includes four stages as illustrated in Figure 4, which unfold into the nine steps listed below:

1) Select the desired vault type.
2) Decompose the chosen geometry into printable patches.
3) Slice the patches into layers.
4) Run a systematic sensitivity analysis.
5) Determine transition strategies between corbelling and inclined slicing techniques.
6) Define structure thickness, filament count, and infill pattern.
7) Generate a continuous toolpath for 3D printing.
8) Optimize and simulate the designed toolpath.
9) Verify printability of the structure. If it is printable, proceed to the printing step, and if not, return to the decomposition step or adjust the geometry from the beginning.

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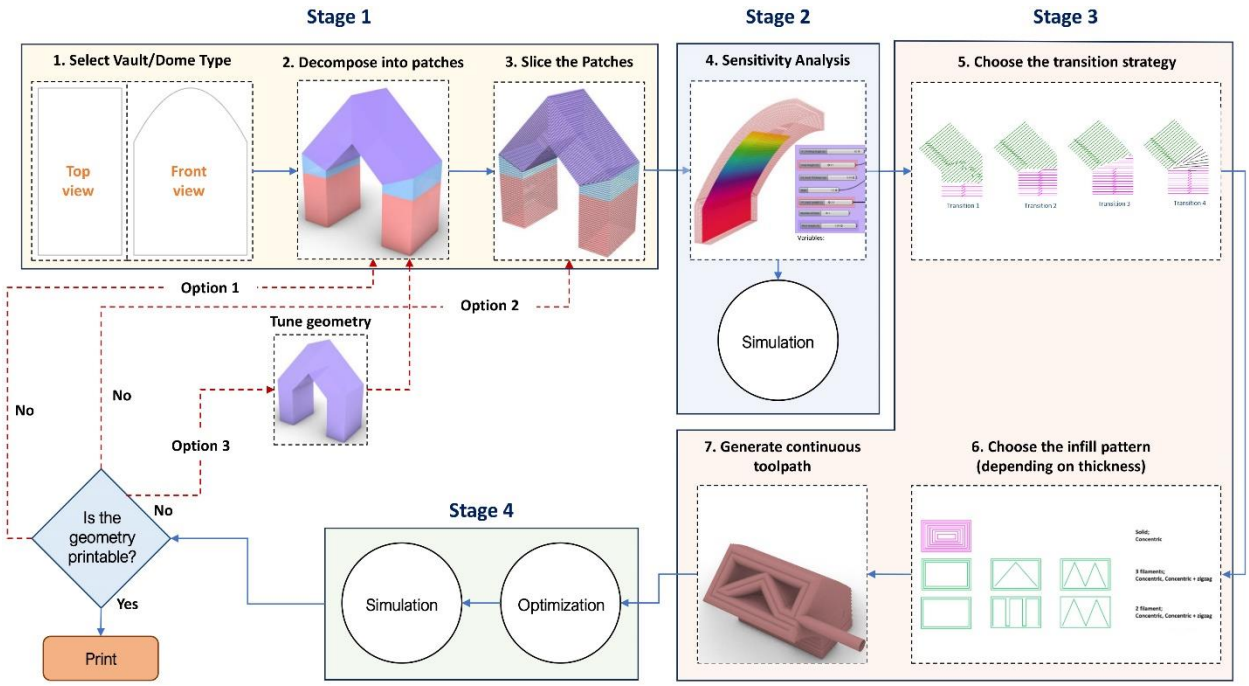

Figure 4. Comprehensive research workflow.

Stage 1 was addressed in Tabassum et al. 2023 and is summarized below, as a segway to describe the current work, focused on Stage 2.

STAGE 1: FORM GENERATION AND DECOMPOSITION

Stage 1 involved the formulation of a shape grammar to generate and decompose the forms of enclosures. By drawing inspiration from historical structures, it was proposed a multi-directional printing approach, integrating corbelling, radial, and inclined slicing techniques for toolpath design. The objective was to establish shape grammar rules to break down enclosures into printable patches, design the corresponding toolpaths using various slicing techniques, and validate the effectiveness of this approach by physically fabricating prototypes. The Stage 1 steps are summarized below and illustrated in Figure 5:

1. **Selection of an enclosure type:** A barrel vault was selected to carry out preliminary experiments.
2. **Form Generation:** A shape grammar was developed to parametrically encode the form of the selected vault.
3. **Form Decomposition:** The form generation shape grammar was complemented with rules for decomposing the overall shape of the vault into parts suitable for 3D printing. This decomposition shape grammar includes rules to decompose the form of the vault into patches, slice patches into layers, and decompose layers into filaments.
4. **Validation:** The form generation and form decomposition shape grammars were validated by the successful 3D printing of part of a barrel vault, featuring a reduced overhang angle of 45 degrees.

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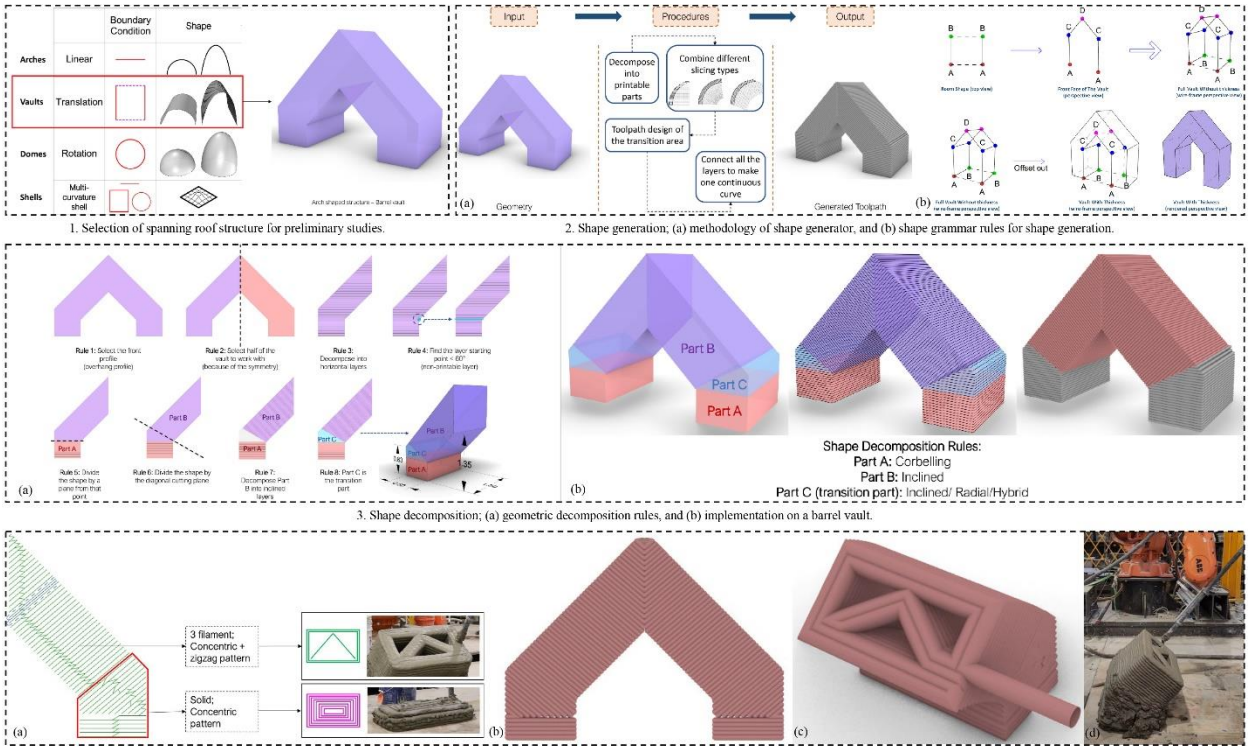

Figure 5. Detailed steps of Stage 1 investigations (Tabassum et al. 2023).

STAGE 2: SENSITIVITY ANALYSIS

Test prints in Stage 1 demonstrated the feasibility of combining corbelling and inclined slicing techniques, allowing for an overhang angle of 45 degrees. However, conducting numerous test prints to gather the data required to fully validate the research hypothesis is impractical because of the time and material resources it would consume. Therefore, an experimental program, termed sensitivity analysis, was devised to understand the impact of geometric variables on the buildability and printability of the vault. This analysis included the generation of barrel vaults with different geometric variable values, their decomposition into toolpaths using different slicing strategies, and then the verification of their printability via the simulator of structural behavior during printing. The workflow of this stage is outlined below.

1. Sensitivity Analysis Workflow:

This step encompasses the following sub steps (Figure 6):

1) Select the vault type for simulation.
2) Determine the variables to manipulate.
3) Conduct simulations to determine maximum the number of printable layers.
4) Analyze collapse modes and document the findings.
5) Select the optimal printing options.

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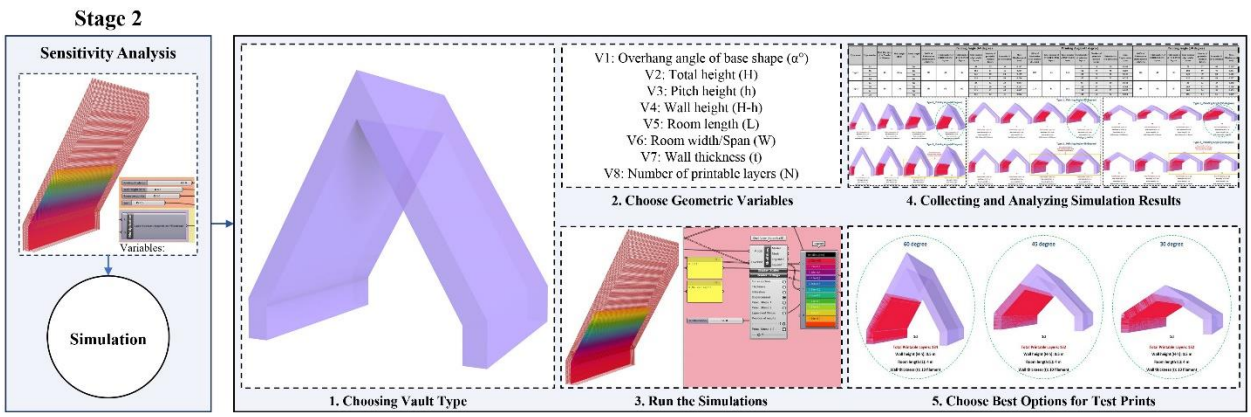

Figure 6. Workflow of sensitivity analysis.

2. Simulation Process:

The geometric variables are listed in Figure 7. Among these variables, two were kept constant (V1-overhang angle, and V6-wall thickness), three were selected as independent variables (V4-wall height, V5-room length, and V7-wall thickness), and three are dependent variables (V2-total height, V3-pitch height, and V8-number of printable layers).

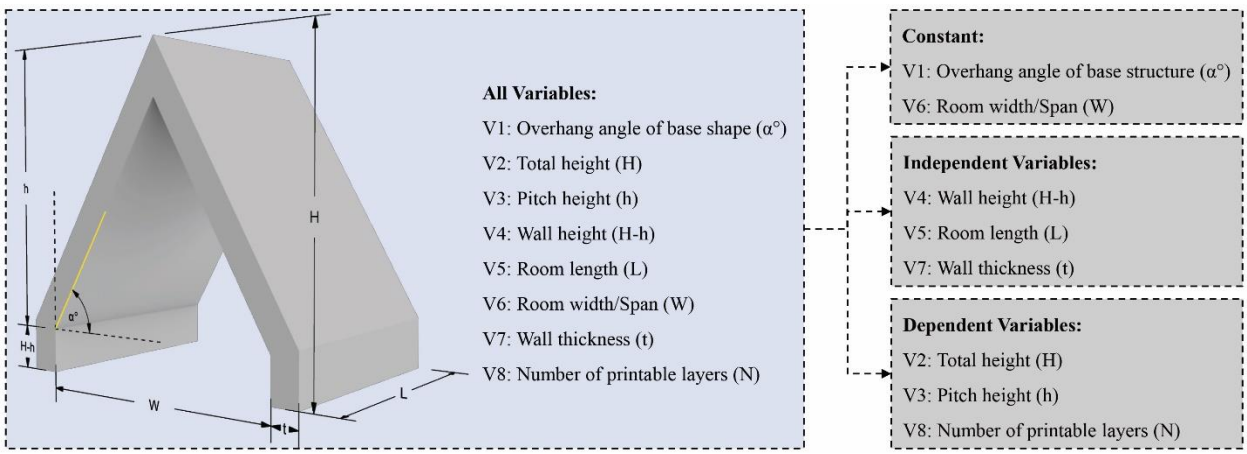

Figure 7. The geometric variables of the barrel vault.

Wall thickness (t) is directly related to the number of filaments (*Nf*), where modifying the number of filaments automatically adjusts the thickness of the geometry, as expressed by the equation:

$$t = Nf \times Wf \tag{1}$$

where *Wf* is the filament width. The relationship between wall thickness, the number of filaments, and the constant filament width is represented in Figure 8 for *Nf* = 10 and *Wf* = 32 mm.

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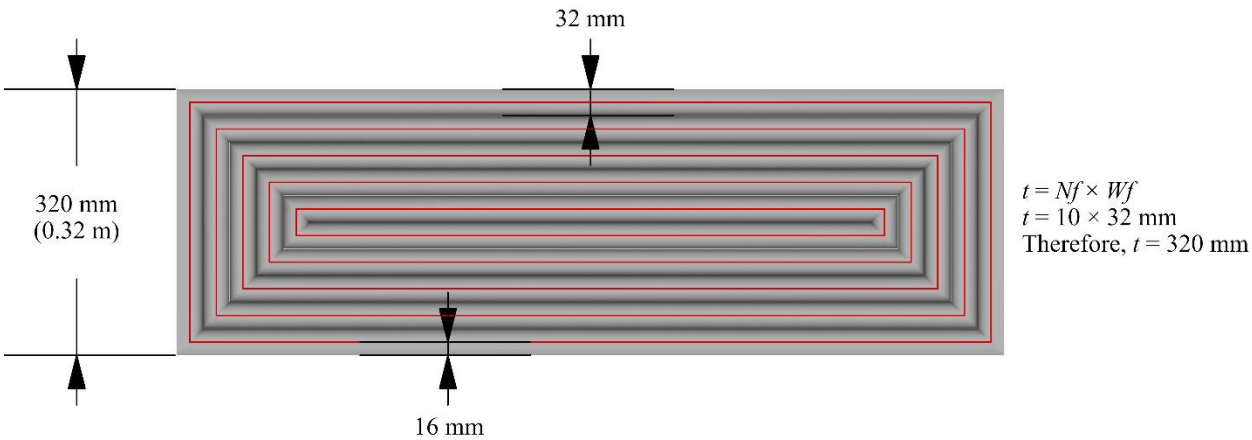

Figure 8. Relationship between wall thickness (t), the number of filaments (*Nf*), and the constant filament width (*Wf*).

In the sensitivity analysis, variables V4, V5, and V7 were manipulated to determine the total number of printable layers. Another critical factor was the angle of printing in inclined slicing, with angles of 60°, 45°, and 30° being considered. The printing angle denotes the rotation of the nozzle relative to the horizontal plane, with the nozzle remaining perpendicular to the filament plane. Various infill patterns provide different toolpath design options that may be better suited to different printing systems, materials, and geometries (Ashrafi et al. 2022). Some of them were explored in Stage 1 experiments, as depicted in Figure 9 (a). However, in the sensitivity analysis, only the concentric solid infill pattern illustrated in Figure 9 (b) was considered.

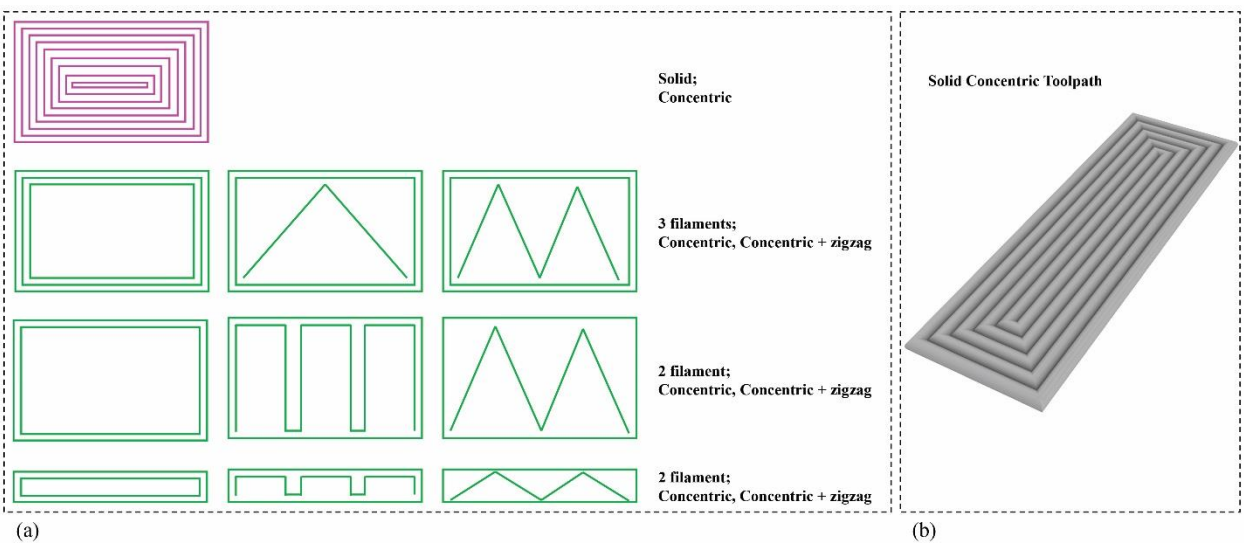

Figure 9. (a) Various types of infill pattern for toolpath design (Tabassum et al. 2023), (b) Solid concentric infill pattern for sensitivity analysis.

The selected combined slicing technique included corbelling for the bottom part and inclined for the top part, as illustrated in Figure 10. The structural performance simulation algorithm developed by Duarte et al. 2023 to verify printability was extended by incorporating inclined slicing, as it

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initially only included corbel slicing. This extension enables the prediction of collapses in structures that incorporate a combination of both corbelling and inclined slicing techniques.

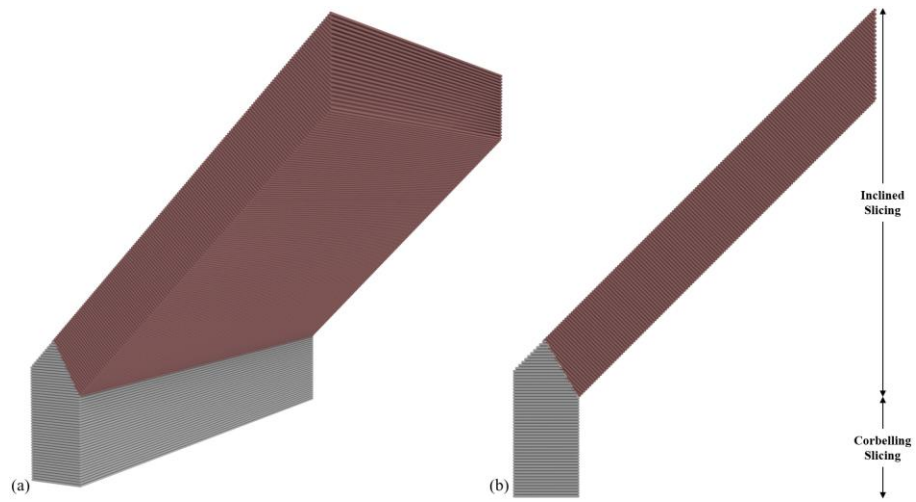

Figure 10. Toolpath with combined slicing: (a) perspective and (b) front view.

Figures 11, 12, and 13 show the results for vault configurations with 60°, 45°, and 30°. The simulation protocol is summarized as follows:

- Manipulation of wall height (*H-h*), room length (*L*), and wall thickness (*t*). The parameters are set as follows:

  H-h = 0.5 m and 1 m
  L = {1, 2, 3, 4} m
  t = {2, 4, 6, 8, 10} filaments

- Changing these three variables directly affects total height (*H*) and the number of printable layers (*N*). For example, when *H-h* = 0.5 m, *L* = 4 m, and *t* = 10, *N* reaches its highest value, which is 134 layers until collapse.

3. Evaluation of the Simulation Outcomes:

a) The analysis of Figures 11a, 12a, and 13a reveals that 1d from Type 1 yielded the highest number of filaments. This configuration is distinguished by shorter wall height and longer room length.
b) While the number of printable layers appears to be nearly the same across different angles, a closer examination of the percentage of fabrication reveals that the 30° printing angle exhibits the highest level of printability, as the number of printable layers corresponds to 91% of the total number of layers that would be necessary to print the entire vault. In this case, 132 layers (91%) out of 146 layers (100%) are printable.
c) A consistent trend is notably observed across all printing angles in Type 2, particularly in configurations 2c and 2d; despite variations in room length, these configurations consistently lead to the same number of printable layers.
d) The highest number of printable layers for 60°, 45°, and 30° are 134, 132, and 132, respectively (Figure 14).

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(a)

Type name	Type number	Wall Thickness (t) or Number of Filament	Wall height (H-h)	Room length (L)	Printing Angle (60 degree)						
					100% of Fabrication (total number of layers)	Total number of CORBELLING layers	Total number of INCLINED layers	Total number of printable layers (N)	Number of printable inclined layers	Printable % of Fabrication	Max. Displacement (cm)
Type 1	1a	10	0.5m	1m	227	43	184	68	25	30	0.117
	1b			2m				104	61	46	0.307
	1c			3m				123	80	54	0.159
	1d			4m				134	91	59	0.236
Type 2	2a	10	1m	1m	253	51	202	83	32	33	0.034
	2b			2m				111	60	44	0.243
	2c			3m				113	62	45	0.062
	2d			4m				113	62	45	0.062

(b)

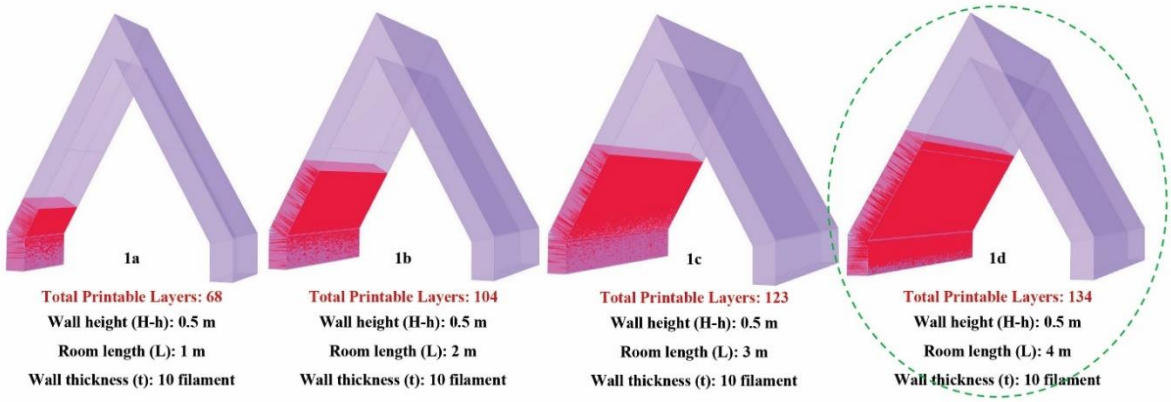

Figure 11. Simulation results with a 60-degree printing angle: (a) table relating geometric variables and outcomes and (b) visualization results for Type 1, highlighting the best outcome.

(a)

Type name	Type number	Wall Thickness (t) or Number of Filament	Wall height (H-h)	Room length (L)	Printing Angle (45 degree)						
					100% of Fabrication (total number of layers)	Total number of CORBELLING layers	Total number of INCLINED layers	Total number of printable layers (N)	Number of printable inclined layers	Printable % of Fabrication	Max. Displacement (cm)
Type 1	1a	10	0.5m	1m	187	43	143	71	28	38	0.148
	1b			2m				102	59	55	0.287
	1c			3m				121	78	65	0.149
	1d			4m				132	89	71	0.222
Type 2	2a	10	1m	1m	213	51	162	85	34	40	0.042
	2b			2m				108	57	51	0.204
	2c			3m				110	61	52	0.052
	2d			4m				110	61	52	0.052

(b)

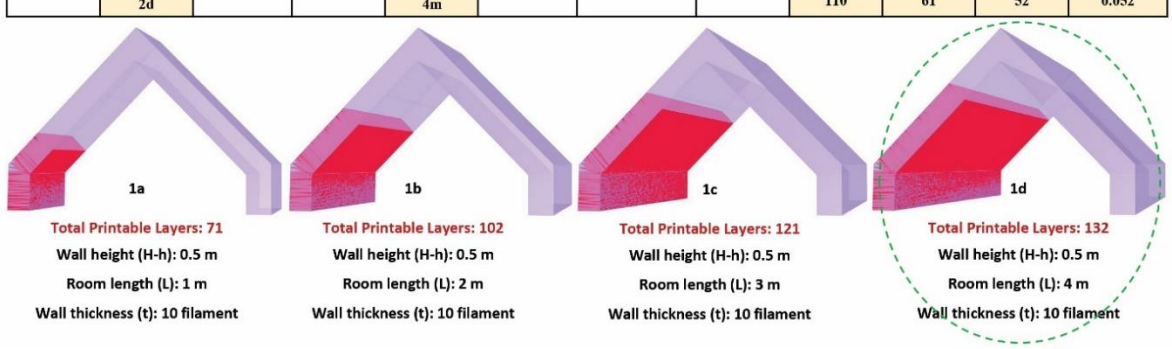

Figure 12. Simulation results with a 45-degree printing angle: (a) table relating geometric variables and outcomes and (b) visualization results for Type 1, highlighting the best outcome.

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(a)

Type name	Type number	Wall Thickness (t) or Number of Filament	Wall height (H-h)	Room length (L)	Printing Angle (30 degree)						
					100% of Fabrication (total number of layers)	Total number of CORBELLING layers	Total number of INCLINED layers	Total number of printable layers (N)	Number of printable inclined layers	Printable % of Fabrication	Max. Displacement (cm)
Type 1	1a	10	0.5m	1m	146	43	103	70	27	48	0.145
	1b			2m				102	59	70	0.304
	1c			3m				122	79	84	0.165
	1d			4m				132	89	91	0.237
Type 2	2a	10	1m	1m	188	51	137	84	33	45	0.038
	2b			2m				103	52	55	0.156
	2c			3m				105	54	56	0.039
	2d			4m				105	54	56	0.039

(b)

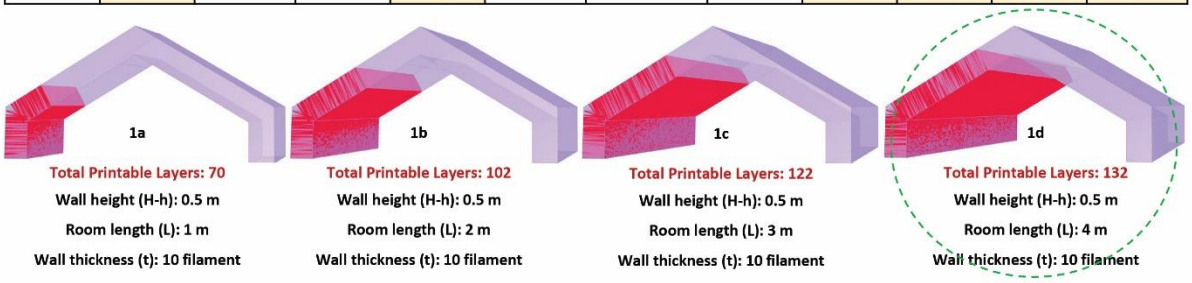

Figure 13. Simulation results with a 30-degree printing angle: (a) table relating geometric variables and outcomes and (b) visualization results for Type 1, highlighting the best outcome.

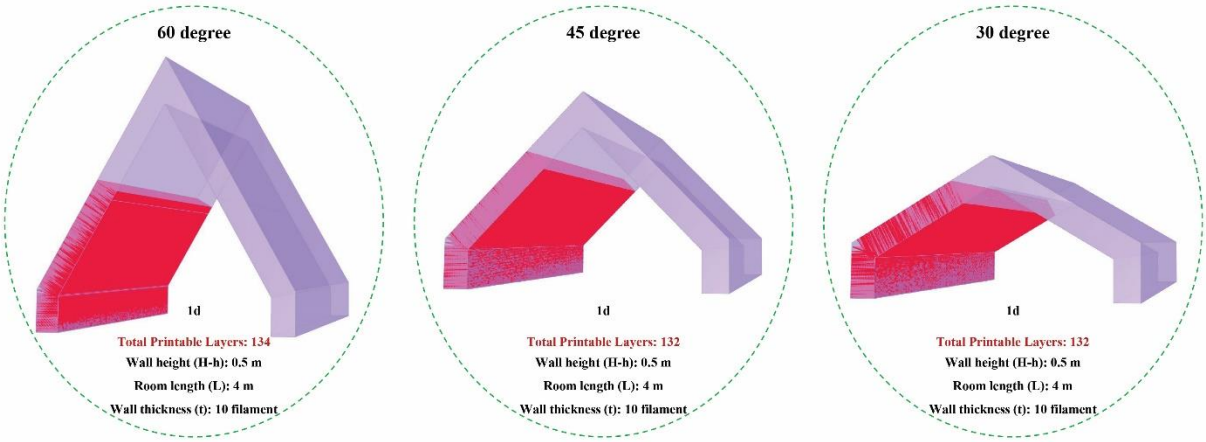

Figure 14. Best results for each printing angle.

4. Findings:

The results of the sensitivity analysis reveal the following findings:

- In general, longer room lengths contribute to increased interlayer printing time and a higher potential for printing more layers. However, this relationship has a limit; for instance, when the room dimensions exceed wall height = 1 m and room length = 3 m, the number of printable layers remains constant.
- To achieve more printable layers, the tradeoffs are twofold:
    a) Decreasing the wall height *H-h* while increasing the room length *L*.
    b) Increasing the room length *L* while decreasing the printing angle.

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- A greater wall thickness (*t*) leads to a greater number of printable layers (*N*).
- Figure 16 shows that a greater number of printable layers (*N*) is possible by lowering the printing angle.

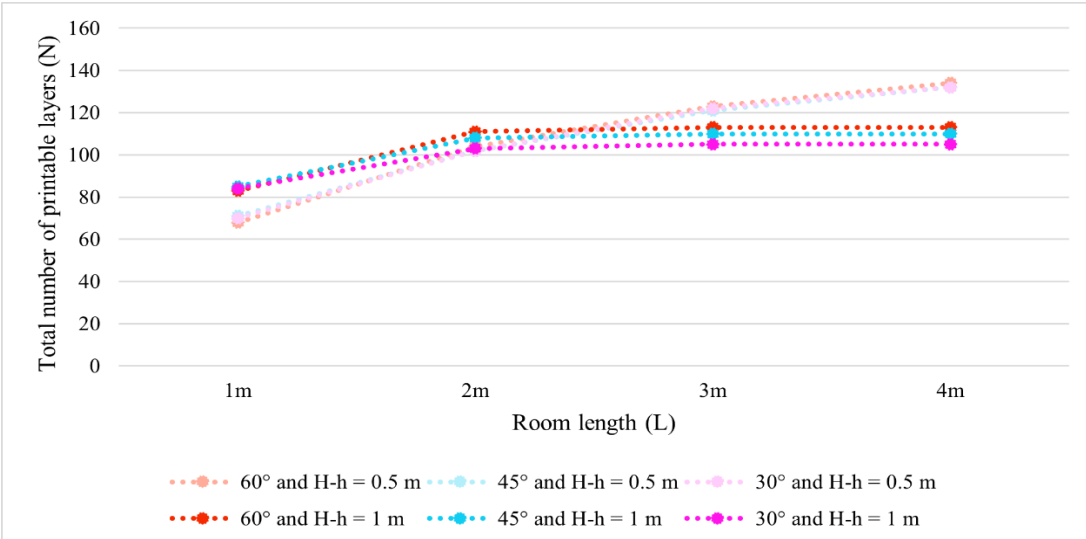

Figure 15. Chart showing the comparison of the simulation results for different room lengths and printing angles.

CONCLUSION

This sensitivity analysis represents an intermediate stage within a broader research agenda, playing a crucial role in comprehending the intricacies of the 3D printing process. The focal point here is a nuanced understanding of how geometric variables exert their influence on the fundamental concept of buildability. Sensitivity analysis serves as a valuable tool in allowing us to perceive patterns that may lead to structural collapse. By manipulating the designs of enclosures based on these observations, we can thoughtfully select prototypes for further validation through printing.

Moving forward, our immediate objective involves validating the simulation outcomes through a series of selected test prints, as outlined in Figure 11. Then, we will analyze the collapse modes of the test prints and investigate any discrepancies with the simulation tool, aiming to understand the reasons behind them. Additionally, we plan to simulate the structural behavior of a vault enclosed by walls on both ends. This comparative analysis will help us understand how additional structures or design elements may impact structural behavior of vaults. This iterative process serves as a crucial bridge, connecting the virtual world of simulations with the tangible reality of 3D printing scenarios, enriching our understanding, and paving the way for developing a tool to assist in the design of enclosures for 3D printing.

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REFERENCES

Antony, Florian, Rainer Grießhammer, Thomas Speck, and Olga Speck. 2014. "Sustainability Assessment of a Lightweight Biomimetic Ceiling Structure." *Bioinspiration & Biomimetics* 9 (1): 016013. https://doi.org/10.1088/1748-3182/9/1/016013.

Ashrafi, Negar, Shadi Nazarian, Nicholas Meisel, and Jose Pinto Duarte. 2022. "A Grammar-Based Algorithm for Toolpath Generation: Compensating for Material Deformation in the Additive Manufacturing of Concrete." *Additive Manufacturing* 55 (July): 102803. https://doi.org/10.1016/j.addma.2022.102803.

Carneau, Paul, Romain Mesnil, Nicolas Roussel, and Olivier Baverel. 2020. "Additive Manufacturing of Cantilever - From Masonry to Concrete 3D Printing." *Automation in Construction* 116 (August): 103184. https://doi.org/10.1016/j.autcon.2020.103184.

Duarte, Gonçalo, Nathan Brown, Ali Memari, and José Pinto Duarte. 2021. "Learning from Historical Structures under Compression for Concrete 3D Printing Construction." *Journal of Building Engineering* 43 (November): 103009. https://doi.org/10.1016/j.jobe.2021.103009.

Duarte, Gonçalo, José Pinto Duarte, Ali Memari, Nathan Brown, and Juan Pablo Gevaudan. 2023. "Towards a Model for Structural Performance in Concrete Printing Based on Buildability and Toolpath Design." *Journal of Building Engineering* 69 (June): 106325. https://doi.org/10.1016/j.jobe.2023.106325.

Kowsky, Joel. 2020. "3D-Printed Habitat Challenge."

Motamedi, Mahan, Romain Mesnil, Robin Oval, and Olivier Baverel. 2022. "Scaffold-Free 3D Printing of Shells: Introduction to Patching Grammar." *Automation in Construction* 139 (July): 104306. https://doi.org/10.1016/j.autcon.2022.104306.

Muthumanickam, Naveen Kumar, Keunhyoung Park, Jose Duarte, Shadi Nazarian, Ali Memari, and Sven Bilén. 2020. *BIM for Parametric Problem Formulation, Optioneering, And 4D Simulation Of 3D-Printed Martian Habitat: A Case Study Of NASA's 3D Printed Habitat Challenge*.

Tabassum, Nusrat, Jose Duarte, and Shadi Nazarian. 2023. "Advancing 3d Concrete Printing for Affordable Housing: A Shape Grammar-Based Approach To Print Spanning Roof Structures."

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